SEQUENCE 13

Synthesis problems, reasoning and fluency

30 March–9 April 2027

13.1 Implication, converse and counterexample

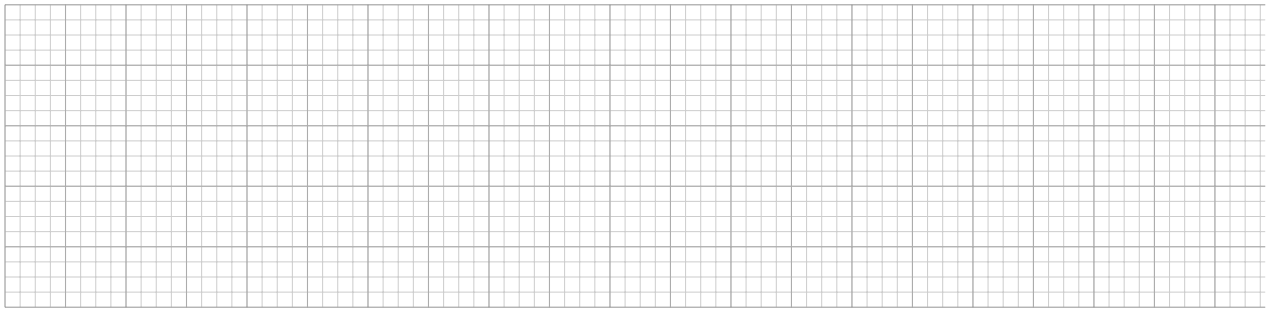
Key point · Tools to use in context. "If P, then Q" is an implication $P \Rightarrow Q$. Its converse is $Q \Rightarrow P$; it is not automatically true. An equivalence $P \iff Q$ requires both implications. A counterexample is enough to refute a universal statement; a few examples are not enough to prove it. The contrapositive not $Q \Rightarrow$ not P is equivalent to $P \Rightarrow Q$.

Example 1. For an integer n: "If n is a multiple of 4, then n is even." Prove the implication, write its converse and its contrapositive, then say which are true.

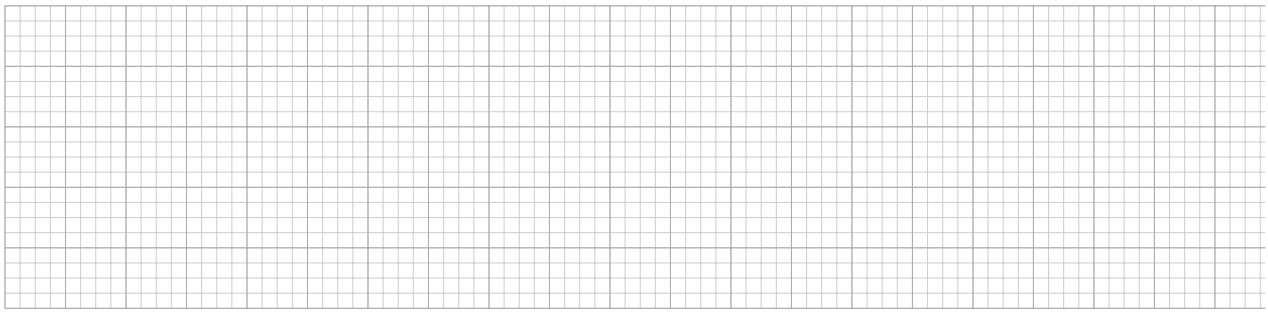
Example 2. For $x \in \mathbb{R}$, study "$x = 3 \Rightarrow x^2 = 9$" and its converse. Give a correct equivalence.

13.2 Negation, case analysis and proof

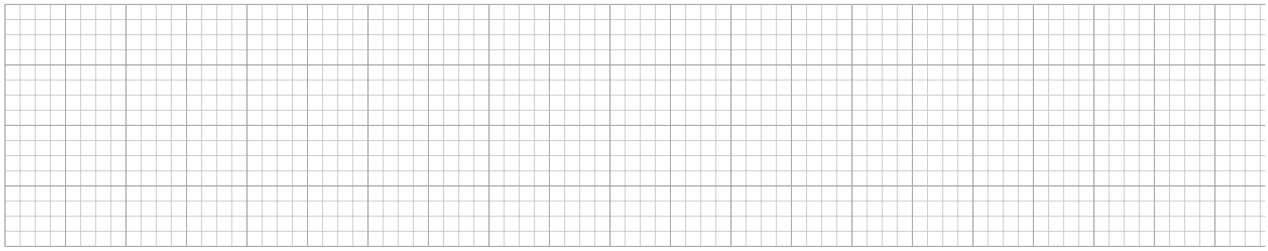
Example 3. Negate correctly: "$x > 2$"; "$x \in [1, 4]$"; "All students play sport"; "A and B both occur".



Example 4. Show by case analysis that $n(n+1)$ is even for every integer n. Then show that n^2+n+1 is odd.



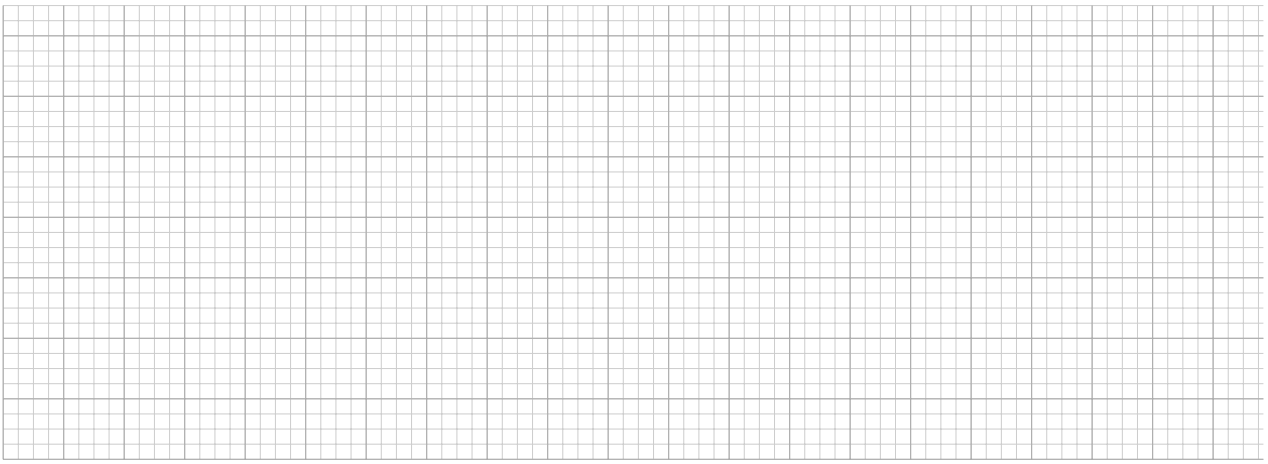
Example 5. A student writes "n^2 even implies n even, because I checked it for $n = 2, 4, 6$". Replace this justification with a proof.

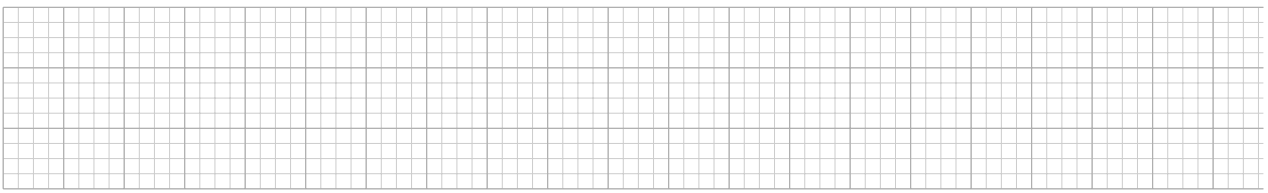


13.3 Combining algebra, functions and percentages

Example 6 · Original situation. Two fictional rental rates are offered for x hours, with $x \in [0, 8]$: $A(x) = 3000 + 2000x$ Ar and $B(x) = 3500x$ Ar.

1. Calculate the costs for 1 h and 4 h.
2. Find where the rates are equal, then the durations for which A is cheaper.
3. A 10% discount is granted on the whole of rate A. Give the new rate A_r and the new comparison threshold.



```
def tarif_moins_cher(x):
    a = 3000 + 2000*x
    b = 3500*x
    if a < b:
        return "A"
    elif b < a:
        return "B"
    else:
        return "egalite"
```

Example 7. Predict the results for $x = 1, 2, 4$. Modify the line that computes a to take the discount into account.